

roomscan — technical report

Phone capture → dimensioned, stitched floor plan with damage scope. All numbers are regenerated by python bench/run\_benchmark.py and written into this report by python bench/make\_report.py (run of 2026-10-04 12:20:54).

0. Scope and the data constraint

We were given three Stray Scanner captures of one flat (LiDAR depth, ARKit poses, intrinsics, RGB video) and no iPhone, no access to the property and no laser. Consequences, stated once and honoured throughout: - **No ground truth**. LiDAR numbers are reported as *repeatability* between three captures of the same rooms (proven to be one flat by wall-map correlation, DECISIONS D6); video/photo numbers are reported against the LiDAR tier of the same capture. Nothing is called “accuracy” that is really agreement. - **Thin tiers are derived**. The video tier gets only rgb.mp4 (upright, metadata stripped); the photo tier gets 2–8 EXIF-free stills per room picked the way the protocol tells a person to shoot. - **Damage is staged digitally** onto real wall planes and re-rendered through the capture’s own poses (§6). - **Head-to-head** with a consumer app needs that app run in the same rooms: not possible here (§8).

1. Architecture

capture → tier front-end → Scene → layout → rooms/walls/openings → damage → JSON + plan  
lidar | video | photo (points, oriented normals, camera views, error model)

Every tier only has to produce a Scene; everything downstream is shared code, which is what makes “same output contract from every tier” true by construction (scene.py, analyze.py, export.py). - **Frame**: gravity from ARKit (LiDAR) or from the floor normal (video/photo); Manhattan yaw from the circular histogram of wall normals; plan coordinates (u, v, h above floor). - **Free space**: per view, a 2-D visibility fan (camera → farthest return per 1° azimuth); union over views. - **Walls**: cells whose vertical points span > 0.9 m of height; Manhattan wall-face segments by 1-D peak finding. - **Rooms**: interior = free – walls – doorway closures; watershed on the distance transform seeded by cores that survive a 0.42 m erosion; room geometry re-snapped to the wall-line arrangement (D4, D13, fix loop). - **Wall faces**: for each polygon edge, the outermost strong, floor-reaching, room-facing surface layer within 0.5 m, fitted from raw points (robust median), never from the 2 cm raster. - **Openings**: 2 cm bins along each face; a run with no surface points in the door band and see-through evidence beyond the wall; widths from jamb points (sub-bin). A gap without see-through is “wall not observed”, never an opening — that is the phantom-opening guard. - **Ceiling**: highest strong horizontal layer inside the room polygon, plane-refined; if the capture never looked up, the value is a *prior* anchored at the highest observed wall point and marked observed: false.

2. Tier design and device matrix

| tier  | front-end                                                                                                                                                                     | scale source                                 | intervals (1σ model, tiers/*.py)                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------|
| LiDAR | Stray loader; RGB/depth frame offset estimated (D9); keyframes 10 cm / 8°                                                                                                     | LiDAR                                        | face 4 mm, scale 0.3%, drift 0.4 mm per m walked |
| video | KLT chains at 10 fps between depth keyframes, PnP-RANSAC with metric mono-depth, ICP / constant-velocity fallback; focal from Manhattan vanishing points (536 vs 533 px true) | mono-depth (Depth Anything V2 metric-indoor) | face 12 mm, scale 2%, drift 2 mm/m               |
| photo | per photo: metric depth → gravity + Manhattan frame; 4 yaws × 2-D shift solved by FFT top-view correlation; rooms joined through doorway shots                                | mono-depth                                   | face 25 mm, scale 3.5%                           |

Hardware: see docs/DEVICE\_MATRIX.md (LiDAR tier needs a Pro iPhone; video/photo any iPhone 15+; processing on a 2-core CPU laptop).

3. Drift

ARKit is locally excellent but drifts over a multi-room walk. We cut the keyframes into ~1.5 m fragments, add ARKit odometry edges between consecutive fragments and **loop-closure edges** between non-consecutive overlapping fragments from point-to-plane ICP, projected to 4-DoF (yaw + translation; gravity is trusted), and optimise the pose graph (Open3D LM with line-process outlier pruning). Same module serves the video tier.

| capture      | drift correction  | footprint m² | wall sharpness | loop residual (cm) before → after | loop edges kept |
|--------------|-------------------|--------------|----------------|-----------------------------------|-----------------|
| single_room  | on                | 23.12        | 0.6998         | 8.4 → 2.7                         | 4               |
| single_room  | off (ARKit as-is) | 23.86        | 0.6065         | —                                 | —               |
| floor_only   | on                | 63.89        | 0.4847         | 14.5 → 6.1                        | 14              |
| floor_only   | off (ARKit as-is) | 64.69        | 0.4981         | —                                 | —               |
| with_ceiling | on                | 68.03        | 0.4622         | 10.2 → 3.3                        | 68              |
| with_ceiling | off (ARKit as-is) | 71.08        | 0.3528         | —                                 | —               |

Cross-capture effect (floor\_only vs with\_ceiling): registration score 0.329 with correction vs 0.301 without; walls within 3%: 30.4% vs 10.5%; median |ΔL| 24.0 vs 25.2 cm.

4. Error budget

Every Measure carries error\_budget\_1sigma: wall length = face fit (each end) ⊕ sensor face bias ⊕ scale × L ⊕ drift × path; area = Σ (L\_i · σ\_face,i) ⊕ scale ⊕ drift; ceiling = floor-plane fit ⊕ ceiling-plane fit ⊕ sensor ⊕ scale; opening width = jamb spread ⊕ sensor ⊕ scale. Unobserved quantities get priors with observed: false.

5. Results

LiDAR repeatability (same tier, different captures of the same rooms)

| pair                     | room pairs | walls | median  ΔL  cm | within 1 cm/0.5% | within 3% | within 8% | ref in 95% CI | openings matched/missed/phantom | opening ≤ 2 cm |
|--------------------------|------------|-------|----------------|------------------|-----------|-----------|---------------|---------------------------------|----------------|
| floor_only~with_ceiling  | 5          | 23    | 24.0           | 4.3%             | 30.4%     | 47.8%     | 47.8%         | 3/6/7                           | 6.2%           |
| with_ceiling~single_room | 2          | 7     | 25.3           | 28.6%            | 42.9%     | 42.9%     | 42.9%         | 1/4/2                           | 14.3%          |
| floor_only~single_room   | 2          | 6     | 17.55          | 16.7%            | 50.0%     | 50.0%     | 50.0%         | 0/1/3                           | 0.0%           |

Thin tiers vs the LiDAR tier of the same capture

Not run in this benchmark: depth weights not present: video and photo tiers not run. The tiers run end-to-end; their odometry/registration code was exercised in development with LiDAR depth substituted for the model (DECISIONS D14), which is not a reported result.

Ceiling heights (only with\_ceiling observes ceilings)

| room | height m | 95% CI         | σ mm |
|------|----------|----------------|------|
| R1   | 2.455    | [2.437, 2.473] | 9.0  |
| R2   | 3.060    | [3.039, 3.081] | 11.0 |
| R3   | 3.028    | [3.007, 3.049] | 11.0 |

|    |       |                |     |
|----|-------|----------------|-----|
| R4 | 2.361 | [2.343, 2.379] | 9.0 |
| R5 | 2.397 | [2.379, 2.415] | 9.0 |

Repeat spread cannot be measured (no second capture looks at the ceilings); the gate row is therefore *unverified*, not passed. Within-capture plane-fit  $\sigma$  is shown.

**Damage (digitally staged, single\_room\_staged)**

recall 0.00 (0/2), false positives 0; clean capture false positives 1.

**Timing (2-core CPU, seconds)**

| run                | reconstruct | layout | damage | total |
|--------------------|-------------|--------|--------|-------|
| lidar/single_room  | 33.8        | 1.5    | 84.5   | 120.3 |
| lidar/floor_only   | 87.5        | 6.7    | 72.1   | 166.6 |
| lidar/with_ceiling | 298.9       | 11.9   | 124.4  | 435.4 |

**6. Calibration**

Uncalibrated (sensor/fit/scale/drift budget only), 59% of repeat-capture wall differences fell inside the combined 95% intervals — overconfident. `bench/calibrate.py` fits a per-face residual on repeat captures (gross mismatches > 30 cm — a different wall, not noise — excluded: 14 of 36) and validates leave-one-pair-out: held-out floor\_only~with\_ceiling: 100% (n=13); held-out floor\_only~single\_room: 75% (n=4); held-out with\_ceiling~single\_room: 100% (n=5). Fitted per-face residual 6.5 cm (ambiguity weight 0.0). The intervals in every JSON are therefore the repeatability we can demonstrate, not the sensor spec. Video/photo intervals use tier error models (scale 2% / 3.5%) that are priors until their benchmark rows exist.

**7. Fix loop**

See docs/FIX\_L00P.md (declaration committed before the fix; before/after regenerable with `bash bench/fix_loop/run.sh before|after`). Worst gate: LiDAR repeatability, 0% of walls within 1 cm / 0.5%. Iteration 1 (capture-invariant room splitting at doorways): 0 → 16.7% / 23.1% on two pairs; median-error prediction badly wrong (post-mortem: a second, systematic cause). Iteration 2 (furniture front vs wall: camera-oriented normals, outermost floor-reaching face): median  $|\Delta L|$  24.9 → 9.2 cm, within-3% 27.8 → 31.6%, but the declared 1 cm metric fell to 5.3% — the remaining error is within-capture drift duplicates (6–10 cm double walls), now attacked by the re-tuned drift correction (D16). Gate still fails; the report says why.

**8. Head-to-head**

Not producible from the sample data: the comparison needs a consumer app (e.g. magicplan, Polycam) run *in the same rooms*. The pipeline side is ready (`bench/compare.py` compares any two plans dimension by dimension; an app export converted to our JSON drops straight in).

**9. Known failure modes (and what the output does about them)**

- **Mirrors** (bathroom): LiDAR returns the mirrored room behind the glass → phantom free space; damage orthomosaics on that wall are reflections. Openings found on a wall are not filtered for mirror symmetry yet.
- **Glass shower screens / glass doors**: partly transparent to LiDAR; walls behind them are seen through reflections → damage detection unreliable there (staged stain missed behind the screen, §5).
- **Wet-look / glossy floors**: specular dropouts in depth; floors are not assessed for damage.
- **Low light**: LiDAR unaffected; mono-depth and KLT degrade — the video tier falls back to ICP/constant velocity (counted in `capture.odometry`) and its intervals are scale-dominated anyway.
- **Furniture against walls**: a face that stops short of the floor is not accepted as the wall when a floor-reaching face exists behind it; a full-height wardrobe is indistinguishable from a wall.
- **Open-plan junctions** (> 1.25 m): consistently merged into one room — a homeowner may draw two.
- **Ceiling never looked at**: reported as a prior, not a measurement.